QG-CRM: Ultraviolet Completion of the Curvature Relaxation Model via Quantum Quadratic Gravity

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Papers I–IV of the Curvature Relaxation Model (CRM) series demonstrated that late-time cosmic acceleration and galactic dynamics can be explained geometrically via a saturation mechanism $\Omega_\Phi(a) = \Phi_0 \tanh(k(a - a_{\rm trans}))$, eliminating both dark energy and particulate dark matter. Paper V elevated this profile to a *universal normal form* via the Saturation Theorem, with the open question: *which ultraviolet completion selects k and Φ_0?* This paper answers that question. We show that the γR^2 term in the CRM Lagrangian (Paper III) admits a natural ultraviolet completion via asymptotically free quantum quadratic gravity (QQG), following the framework of Liu, Quintin & Afshordi [6]. In the UV regime, 1-loop renormalization group running of the R^2 coupling $\xi(\mu)$ dynamically generates slow-roll inflation without an inflaton field. We verify that the four axioms of the Saturation Theorem are satisfied by the QQG renormalization group flow, establishing the tanh profile as the unique infrared attractor. The resulting framework—*QG-CRM*—provides a complete geometric narrative: (i) asymptotically free R^2 gravity drives inflation in the UV; (ii) saturation dynamics governs the graceful exit and the emergence of general relativity; (iii) curvature relaxation produces geometric dark matter and late-time acceleration in the IR. We derive the explicit parameter mapping from QQG couplings (λ_0, $\mathcal{N}$, μ_0) to CRM observables (k, Φ_0, $a_{\rm trans}$). The tensor-induced inflation scenario of Bertacca et al. [7] is classified as a structurally distinct limiting case that falls outside the Saturation Theorem axioms. Observational predictions include $n_s \sim 1 - 4/(3N)$, $r \gtrsim 0.01$, $w(z) < -1$ at late times, and the MOND consistency relation $a_0 = cH_0/(2\pi)$—all testable with Stage IV CMB experiments, Euclid, and Roman.

I. INTRODUCTION

The Curvature Relaxation Model (CRM) provides a geometric alternative to the standard ΛCDM cosmology. Over five companion papers, the framework has been developed from game-theoretic axioms (Paper I [1]), through a unified treatment of dark energy and MOND-like dark matter (Paper II [2]), to microscopic foundations via an $R + \gamma R^2$ Lagrangian with Pöschl-Teller saturation (Paper III [3]), the galactic-cosmological nexus via a vector sector (Paper IV [4]), and the Saturation Theorem establishing tanh as the universal normal form of capacity-limited cooperative relaxation (Paper V [5]).

The central phenomenological result—the saturation ODE $dX/da = k(1-X^2)$ and its unique solution $X(a) = \tanh(k(a-a_0))$—has been tested against Pantheon+ supernovae [29], Planck CMB observables [31], DESI BAO measurements [32], and SPARC rotation curves (Papers I–IV), achieving $\Delta\chi^2 = -5.5$ vs. ΛCDM with zero early dark energy and only 6 free parameters.

However, the CRM series has so far remained agnostic about the ultraviolet regime. Paper V concluded with an explicit open question:

> "The open question being which UV completion selects k and Φ_0."

This paper answers that question. We identify the γR^2 sector of the CRM Lagrangian with quantum quadratic gravity (QQG), which is asymptotically free in the ultraviolet and whose 1-loop renormalization group (RG) running dynamically generates slow-roll inflation [6]. The Saturation Theorem provides the interface: it constrains the infrared behavior to tanh regardless of UV details, while the UV completion fixes the microscopic parameters.

The resulting framework, which we call *QG-CRM*, provides a complete geometric narrative from the Big Bang to the present epoch—without an inflaton, without a cosmological constant, and without dark matter particles.

We emphasize that this paper establishes *structural necessity*, not numerical closure. The UV completion determines the existence, functional form, and scaling relations of the CRM parameters; their precise numerical values require nonperturbative UV-IR matching analogous to computing hadron masses from the QCD Lagrangian. What we demonstrate is that QG-CRM is the unique consistent framework connecting asymptotically free quadratic gravity to capacity-limited cosmological relaxation.

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II. THE CRM R^2 SECTOR AND THE ULTRAVIOLET REGIME

A. Recap: The CRM Lagrangian

Paper III established the effective CRM Lagrangian [3]:

$$\mathcal{L}_{\rm CRM} = \frac{R}{16\pi G} + \gamma R^2 + \frac{1}{2}(\partial\phi)^2 - V_{\rm PT}(\phi), \qquad (1)$$

where γR^2 generates the geometric "dark matter" contribution (the scalaron), and the Pöschl-Teller potential $V_{\rm PT}(\phi)$ provides the saturation dynamics driving late-time acceleration.

In the very early universe, when curvature is Planckian, the Einstein-Hilbert term $R/(16\pi G) \propto M_{\rm Pl}^2 R$ becomes subdominant relative to the R^2 term. The effective action reduces to

$$S_{\rm UV} \approx -\int d^4x\,\sqrt{-g}\,\frac{R^2}{\xi}, \qquad (2)$$

where $\xi \equiv -1/\gamma$ (following the sign convention of [6]; note that $\gamma > 0$ in Paper III corresponds to $\xi < 0$, i.e., the tachyon-free region that the RG trajectory reaches in the IR). During inflation, $\xi > 0$. This is a scale-invariant theory: any constant-R spacetime—Minkowski, de Sitter, or anti-de Sitter—extremizes the action [10, 11]. The addition of R^2 to the Einstein-Hilbert action is also the starting point of Starobinsky inflation [9], while the full quadratic gravity action including C^2 is known to be renormalizable [17].

B. From scale invariance to inflation

Scale invariance is broken by quantum effects. The renormalization group running of $\xi(\mu)$ introduces a physical scale into the theory, analogous to the quantum conformal anomaly [12, 13]. This is precisely the mechanism that transforms exact de Sitter into quasi-de Sitter, generating slow-roll inflation without introducing an inflaton field.

On a homogeneous and isotropic FLRW background, the Weyl tensor C^2 vanishes identically. Therefore, the full quadratic gravity action

$$S_{\rm QG} = -\int d^4x\,\sqrt{-g}\left(\frac{R^2}{\xi} + \frac{C^2}{2\lambda}\right) \qquad (3)$$

reduces to pure R^2 gravity for the cosmological background, and the inflationary dynamics is entirely determined by the running of ξ.

III. ULTRAVIOLET COMPLETION VIA QUANTUM QUADRATIC GRAVITY

A. The renormalization group flow

Following [6, 8], the 1-loop beta functions of the physical running couplings are

$$\beta_\xi = \frac{d\xi}{d\ln\mu} = -\frac{1}{(4\pi)^2}\,\frac{\xi^2 - 36\lambda\xi - 2520\lambda^2}{36}, \qquad (4)$$

$$\beta_\lambda = \frac{d\lambda}{d\ln\mu} = -\frac{1}{(4\pi)^2}\,\frac{(1617 + 90\mathcal{N})\lambda - 20\xi}{90}\,\lambda, \qquad (5)$$

where μ is the physical RG scale and $\mathcal{N}$ counts the matter field content:

$$\mathcal{N} = \frac{1}{60}N_{\rm scalar} + \frac{1}{5}N_{\rm vector} + \frac{1}{20}N_{\rm fermion}. \qquad (6)$$

B. Asymptotic freedom and the inflationary trajectory

The RG flow (4)–(5) admits solutions that are:

1. **Asymptotically free in the UV**: both ξ and λ vanish at the Gaussian fixed point ($\mu \to \infty$).
2. **Tachyon-free in the IR**: the low-μ basin of attraction has $\lambda > 0$ and $\xi < 0$.

Trajectories reaching the tachyon-free region must first pass through a regime where $\xi > 0$: the coupling grows, reaches a maximum $\xi_{\rm max}$, then decreases and crosses zero at a scale μ_0. Inflation occurs in the regime of *decreasing* ξ, before the tachyon divide is crossed [6].

C. Large-$\mathcal{N}$ solution and the inflation potential

In the large-$\mathcal{N}$ limit, the solution reads [6]:

$$\xi(\mu) \simeq \frac{35\lambda_0^2\ln(\mu/\mu_0)}{8\pi^2[1 + \lambda_{\rm tH}\ln(\mu/\mu_0)]}, \qquad (7)$$

where $\lambda_{\rm tH} \equiv \lambda_0\mathcal{N}/(4\pi)^2$ is the 't Hooft-like coupling. Promoting $\mu = |R|^{1/2}$ and transforming to the Einstein frame yields the scalar potential [6]:

$$V(\varphi) \simeq \frac{35\lambda_0^2\mu_0^4}{128\pi^2\lambda_{\rm tH}}\left(1 - \frac{\sqrt{6}\,\mu_0}{\lambda_{\rm tH}\,\varphi}\right). \qquad (8)$$

This potential has a near-plateau structure, yielding slow-roll inflation with observables [6]:

$$n_s \sim 1 - \frac{4}{3N}, \qquad r \sim \frac{8}{3}\left(\frac{2}{\lambda_{\rm tH}^2 N^4}\right)^{1/3}, \qquad (9)$$

where N is the number of e-folds between horizon exit and the end of inflation.

D. End of inflation and the onset of kination

Inflation ends as the potential steepens; there is no minimum where the inflaton could oscillate and decay. Instead, the field rolls into a kinetic-dominated phase (*kination*). As λ flows beyond λ_0, the 't Hooft coupling $\lambda_{\rm tH} = \lambda\mathcal{N}/(4\pi)^2$ surpasses unity, signaling the strong-coupling regime. General relativity must then emerge as the effective field theory at low energies, and the universe enters its radiation era via reheating [6].

This UV→IR transition is precisely where the CRM's infrared dynamics takes over.

IV. THE SATURATION THEOREM AS UV-IR INTERFACE

A. Axiom verification for QQG

Paper V established four axioms that any quantum gravity theory must satisfy to produce bounded cosmological relaxation [5]. We now verify that each axiom is satisfied by QQG in the cosmological sector.

Axiom A (Finite geometric capacity).: The coupling $\xi(\mu)$ reaches a finite maximum $\xi_{\rm max}$ along the RG trajectory before decreasing. This maximal curvature response is the geometric capacity bound. ✓

Axiom B (Cooperative reinforcement).: The RG running of ξ is self-reinforcing in the inflationary regime: the ξ^2 term in β_ξ generates cooperative growth from the UV fixed point to $\xi_{\rm max}$, followed by cooperative relaxation as ξ decreases. ✓

Axiom C (Monotone cosmic control).: The physical RG scale μ decreases monotonically as the universe expands. During inflation and matter domination, $\mu = |R|^{1/2}$ provides the natural identification. In the radiation era, where the trace $R = 0$ for conformally coupled matter, μ should be identified with H (the Hubble scale) or an alternative curvature invariant such as the Kretschner scalar $K^{1/4}$; all choices agree during slow-roll inflation and are generically equivalent up to $\mathcal{O}(1)$ factors [6].[1] The monotonicity of cosmic expansion ensures that μ decreases in all phases. ✓

[1] The $\mathcal{O}(1)$ equivalence holds because during quasi-de Sitter expansion $R \approx 12H^2$, so $|R|^{1/2} \approx 2\sqrt{3}\,H$ and $K^{1/4} \approx 2\sqrt{3}\,H$. In the radiation era, $R = 0$ but $K = 12H^4 > 0$ and H itself remains monotone. The inflationary predictions depend only on the slow-roll regime where all choices coincide; the radiation-era ambiguity affects at most the end-of-inflation matching at the $\mathcal{O}(1)$ level.

Axiom D (RG composition).: The 1-loop beta functions (4)–(5) define a well-posed composition law: evolving from $\mu_1 \to \mu_2 \to \mu_3$ is equivalent to evolving directly from $\mu_1 \to \mu_3$. This is precisely the semigroup property of the RG flow, satisfying the Abel equation requirement of the Saturation Theorem. ✓

B. Consequence: tanh as the unique IR attractor

Since all four axioms are satisfied, the Saturation Theorem applies: the unique smooth, scale-coherent response function extending regularly to the saturation boundary is

$$S(g) = S_{\rm max}\,\tanh(\alpha\,u(g)), \tag{10}$$

up to reparametrization. In cosmological language, this is the CRM saturation profile $\Omega_\Phi(a) = \Phi_0 \tanh(k(a - a_{\rm trans}))$.

C. Parameter identification

The UV completion determines the reparametrization $u(g)$ and the CRM constants:

- The saturation amplitude Φ_0 is fixed by $\xi_{\rm max}$ and λ_0.
- The relaxation rate k is determined by the slope of $\xi(\mu)$ at the tachyon divide.
- The transition scale $a_{\rm trans}$ corresponds to the scale μ_0 where $\xi = 0$ and GR emerges.

The explicit mapping is derived in Appendix A.

V. FROM INFLATION TO LATE-TIME ACCELERATION: THE QG-CRM NARRATIVE

QG-CRM provides a unified geometric narrative spanning the entire cosmic history:

1. **Phase I: Pure R^2 inflation (UV, $\mu \gg \mu_0$).** The universe begins in asymptotically free quadratic gravity. The R^2 coupling $\xi(\mu)$ runs from zero, reaches $\xi_{\rm max}$, and generates quasi-de Sitter inflation via the logarithmic correction to the R^2 action. No inflaton field exists; the scalaron is purely geometric.

2. **Phase II: Kination and strong coupling ($\mu \sim \mu_0$).** As ξ decreases and crosses zero, the potential steepens, inflation ends, and the field enters kination. The 't Hooft coupling approaches unity, signaling the strong-coupling regime. General relativity emerges as the effective field theory.

3. **Phase III: Radiation and matter era.** Standard hot Big Bang cosmology. The Pöschl-Teller scalar ϕ is dynamically frozen by the trace coupling $\mathcal{F}(T/\rho)$ (Paper III), ensuring BBN protection.

4. **Phase IV: Geometric dark matter ($z \lesssim 10^3$).** The scalaron from the γR^2 term produces scale-dependent gravitational effects that mimic cold dark matter cosmologically (running coupling $\beta_{\rm eff}(a)$, Paper II) and reproduce flat rotation curves via the vector sector (Paper IV).

5. **Phase V: Curvature saturation ($z \lesssim 1$).** The Pöschl-Teller potential activates, the curvature return potential saturates ($\Omega_\Phi \to \Phi_0$), and the universe enters accelerated expansion. The saturation is guaranteed to follow tanh by the Saturation Theorem.

This five-phase narrative is not a collection of separate models stitched together. All phases arise from a *single* geometric degree of freedom—the R^2 scalaron—whose dynamics is governed by renormalization group flow in the UV and saturation dynamics in the IR, with the Saturation Theorem providing the unique bridge.

A. Scale-dependent scalaron mass: one field, many roles

A potential objection is that the scalaron cannot simultaneously drive inflation ($m_s \sim 10^{13}\,$GeV in Starobinsky inflation) and produce geometric dark matter effects ($m_s \sim H_0 \sim 10^{-33}\,$eV at cosmological scales). The resolution lies in the *running* of $\gamma(\mu)$: since $\gamma = -1/\xi$ and $\xi(\mu)$ evolves along the RG trajectory, the scalaron mass

$$m_s^2(\mu) = \frac{1}{6\gamma(\mu)} = -\frac{\xi(\mu)}{6} \tag{11}$$

is not a fixed parameter but a scale-dependent effective quantity. In the UV ($\mu \gg \mu_0$), ξ is large and positive, yielding a heavy scalaron that drives inflationary dynamics. As ξ decreases through the RG flow, the effective mass drops. After the strong-coupling transition ($\mu \sim \mu_0$), the scalaron enters the tachyon-free regime ($\xi < 0$, $\gamma > 0$), where its mass is controlled by the IR value of γ—potentially as light as $m_s \sim H_0$.

The "identity" of the scalaron across all five phases is therefore not a claim that a single mass parameter governs all epochs, but that a single *geometric degree of freedom* with a *scale-dependent effective description* provides the underlying dynamics. This is directly analogous to the QCD gluon: the same gauge field is perturbative at high energies and confining at low energies, with a scale-dependent effective coupling $\alpha_s(\mu)$ that changes by orders of magnitude.

B. The $R^2 \to$ nonlinear $f(R)$ transition and solar system screening

A crucial constraint arises from the companion paper on dark energy [15]: the *Convexity-Screening Incompatibility Theorem* establishes that a strictly convex free-energy functional—as naturally produced by linear $R + \gamma R^2$ gravity—admits a unique global minimum with no secondary minima. Consequently, linear R^2 gravity *cannot* produce chameleon screening: the post-Newtonian parameter evaluates to $\gamma_{\rm PPN} = 1/2$, excluded by Cassini tracking at $> 10^4\sigma$.

This has a direct implication for QG-CRM: the UV action (3) is renormalizably R^2, but the IR effective theory *must* develop nonlinear $f(R)$ structure to achieve solar system compatibility. Concretely, the Hu-Sawicki form [16]

$$f(R) = R - \mu^2 \frac{c_1 (R/\mu^2)^n}{c_2 (R/\mu^2)^n + 1}, \qquad n \geq 1, \tag{12}$$

provides density-dependent screening: the scalaron mass $m_s^2 \propto \rho/M_{\rm Pl}^2$ becomes large in high-density environments (solar system), suppressing the Yukawa fifth force, while remaining light ($m_s \sim H_0$) at cosmological densities where dark energy effects are active.

In QG-CRM, this transition has a natural interpretation as an *irrelevant-operator cascade*. In the UV, the theory is perturbatively R^2 (renormalizable, with only marginal and relevant operators). As the RG flow approaches strong coupling, nonperturbative effects generate an infinite tower of higher-curvature operators:

$$\Gamma_{\rm IR} = \Gamma_{R^2} + \sum_{n\geq 1} \frac{c_n\, R^{n+1}}{\Lambda^{2n}}, \tag{13}$$

where Λ is the confinement scale. The Hu-Sawicki form (12) is then not a fundamental action but the leading *phenomenological resummation* of this operator tower that captures the essential physics: density-dependent scalaron mass and chameleon screening. This is directly analogous to chiral perturbation theory, which parametrizes the low-energy effects of QCD confinement without deriving them from the QCD Lagrangian.

We emphasize that Hu-Sawicki should be understood as an IR effective parametrization of nonperturbative completion, not as a derived action. The specific values of n, c_1, c_2 are constrained by solar system tests and cosmological data, not by the UV theory. What the UV theory guarantees is that *some* nonlinear $f(R)$ structure must emerge at the strong-coupling transition—the Convexity-Screening Theorem makes this a structural necessity, not an optional addition.

C. Connection to the dark energy paper: thermodynamic consistency

The companion FST dark energy paper [15] derives a thermodynamic bound on the effective cosmological con-

stant:

$$\Lambda_{\rm eff} \sim \frac{H_0^2}{\ln(M_{\rm Pl}/H_0)}, \tag{14}$$

yielding $\Lambda_{\rm eff} \approx 3.8 \times 10^{-53}\,{\rm m}^{-2}$, within an order of magnitude of the Planck 2018 value ($1.1 \times 10^{-52}\,{\rm m}^{-2}$). Since the same Lagrangian ($R + \gamma R^2$ + Pöschl-Teller) underlies both works, the thermodynamic bound provides an order-of-magnitude *lower bound* on the vacuum energy density. Note that the CRM saturation amplitude $\Phi_0 \equiv \Omega_\Lambda \approx 0.685$ (the fractional dark energy density parameter) is a different quantity from $\Lambda_{\rm eff}/(3H_0^2)$. The thermodynamic functional selects the vacuum energy *scale*; the saturation amplitude Φ_0 is determined by the full cosmic expansion history. The ratio

$$\frac{\Lambda_{\rm eff}^{\rm thermo}}{3H_0^2} \sim \frac{1}{3\ln(M_{\rm Pl}/H_0)} \approx 0.0024 \tag{15}$$

indicates that the raw thermodynamic bound underestimates Ω_Λ by a factor $\sim 290 \approx 2\ln(M_{\rm Pl}/H_0)$, identifying the RG matching factor needed to close the gap (see Section VIII G).

Furthermore, the DESI-compatible best-fit parameters from [15] ($\kappa = 2.63$, $a_{\rm trans} = 0.326$) provide concrete target values for the QQG→CRM parameter mapping (Appendix A).

A particularly intriguing prospect is that the QQG renormalization group flow may close the principal open problem identified in [15]: the missing factor $\sim 2\ln(M_{\rm Pl}/H_0) \approx 290$. The thermodynamic functional provides one logarithm from the entropy/state-density logic; a second logarithm arises naturally from the RG integration of the vacuum energy running:

$$\Lambda_{\rm eff} \sim \int_{H_0}^{M_{\rm Pl}} \frac{d\ln\mu}{\mu}\,\beta_\Lambda(\mu) \sim \frac{H_0^2\,\ln^2(M_{\rm Pl}/H_0)}{(4\pi)^2}, \tag{16}$$

where the double-logarithm structure is a standard feature of integrated RG flows in quantum field theory. This "double-log" mechanism—one ln from thermodynamics, one from UV→IR integration—would yield $\Phi_0 \sim \ln(M_{\rm Pl}/H_0)/(3\ln(M_{\rm Pl}/H_0)) \times \mathcal{O}(\text{few}) \sim \mathcal{O}(1)$, consistent with $\Omega_\Lambda \approx 0.685$. A rigorous derivation requires explicit evaluation of β_Λ in QQG, which we defer to future work.

VI. RELATION TO TENSOR-INDUCED INFLATION

Bertacca et al. [7] proposed a scenario—building on the foundational work of Matarrese, Mollerach & Bruni [21]—in which scalar perturbations arise purely as a second-order effect from tensor fluctuations in exact de Sitter, with no scalar degrees of freedom at linear order. While this is an elegant construction, it is structurally incompatible with saturation cosmology for the following reasons:

1. **No linear scalar sector.** The scenario explicitly excludes linear scalar perturbations. The CRM, by contrast, requires the scalaron as a dynamical degree of freedom for both the inflationary background and the late-time saturation mechanism.

2. **No saturation dynamics.** Without a scalar order parameter, there is no composable relaxation variable satisfying the Abel equation. The Saturation Theorem is therefore not applicable in this setting, and the tanh normal form cannot be derived from the tensor-only framework.

3. **No global cosmological program.** The tensor-induced scenario addresses inflation and perturbation generation, but does not extend to dark matter, MOND, or late-time acceleration.

However, the tensor-to-scalar sourcing mechanism of [7] may provide a *subleading correction* to the scalaron-dominated power spectrum in QG-CRM. The quantitative assessment of this correction—whether the second-order tensor channel modifies $\mathcal{P}_\zeta(k)$ at the percent level—remains an open question for future work.

Remark 1. *The scenario of [7] predicts a very different tensor-to-scalar ratio ($r \sim 0.0006$, near GUT scale $H_{\rm inf} \sim 5\times10^{-4}\,m_{\rm pl}$) from the QQG prediction ($r \gtrsim 0.01$). Stage IV CMB experiments will decisively distinguish these scenarios.*

VII. OBSERVATIONAL PREDICTIONS OF QG-CRM

QG-CRM makes concrete, falsifiable predictions spanning the entire cosmic history:

A. Inflationary epoch (UV)

We have computed the inflationary observables numerically across the viable parameter space. The results are summarized in Table I.

TABLE I. QG-CRM inflationary predictions for representative parameter values. The observed scalar amplitude $A_s = e^{3.06} \times 10^{-10}$ is used to fix λ_0 for each $(\lambda_{\rm tH}, N)$ pair.

$\lambda_{\rm tH}$	N	n_s	r	λ_0	$\mathcal{N}$
0.3	55	0.9758	0.036	1.1×10^{-4}	4.3×10^5
0.5	55	0.9758	0.025	1.2×10^{-4}	6.6×10^5
1.0	55	0.9758	0.016	1.4×10^{-4}	1.2×10^6
Starobinsky ($N = 55$)		0.9636	0.004	—	—
ACT+DESI (1σ)		0.974 ± 0.003	—	—	—
SPT-3G+DESI (1σ)		0.978 ± 0.004	—	—	—
BICEP/Keck (2σ)		—	< 0.036	—	—

- **Scalar spectral index:** $n_s = 1 - 4/(3N) = 0.9758$ for $N = 55$, lying within 1σ of both ACT+DESI [22] and SPT-3G+DESI [23]. This is significantly larger than Starobinsky's $n_s = 1 - 2/N = 0.964$, which is under tension with recent CMB data.

- **Tensor-to-scalar ratio:** r ranges from 0.016 ($\lambda_{\rm tH} = 1$, at the strong-coupling boundary) to 0.036 ($\lambda_{\rm tH} = 0.3$), with the minimum prediction $r \gtrsim 0.01$. The upper end is at the current BICEP/Keck 2σ bound. LiteBIRD ($\sigma_r \sim 0.001$) and CMB-S4 will probe the entire viable range.

- **Scalar amplitude:** Fixing $A_s = 2.13 \times 10^{-9}$ constrains $\lambda_0 \sim 10^{-4}$ and requires $\mathcal{N} \sim 10^5$–10^6 matter fields in the loops. This large $\mathcal{N}$ is compatible with extensions of the standard model and with holographic cosmology [33].

- **Tensor-sourced correction.** The second-order tensor→scalar contribution of [7] is suppressed by $(H_{\rm inf}/m_{\rm pl})^2 \sim 2.5 \times 10^{-7}$ relative to the scalaron channel, confirming its subleading nature.

B. Late-time cosmology (IR)

Inherited from Papers I–IV:

- **Equation of state and phantom stability:** $w_{\rm eff}(z) < -1$ with $|\Delta w| \approx 0.4$ (phantom-like). Crucially, this phantom phase is *transient*, not asymptotic. As $a \to \infty$: $\Omega_m a^{-3} \to 0$, $\Omega_\Phi \to \Phi_0$, hence $H^2 \to H_0^2 \Phi_0 = \text{const}$ and $\dot{H} \to 0$. The universe approaches a stable de Sitter end state with $w_{\rm eff} \to -1$ from below. No Big Rip, no future singularity—the saturation bound $\Omega_\Phi \leq \Phi_0$ is the structural guarantee.

- **Later acceleration onset:** $z_{\rm acc} = 0.52$ vs. 0.84 (ΛCDM). The extended matter-dominated phase explains JWST "Universe Breakers" at $z > 7$ [24].

- **Modified gravity signatures:** $\mu(k, a) \neq 1$, gravitational slip $\eta \neq 1$, lensing $\Sigma = 1$—distinguishing QG-CRM from ΛCDM. Testable with Euclid and Roman.

- **Gravitational wave speed:** $c_T = c$ (exact), consistent with GW170817 [25].

- **MOND consistency relation:** $a_0 = cH_0/(2\pi)$ [30], linking local galactic dynamics to the cosmological expansion rate.

VIII. DISCUSSION

A. The ontological picture

QG-CRM reduces the material content of cosmology to its minimum:

Spacetime geometry (in five phases) + baryonic matter.

No cosmological constant, no inflaton field, no dark matter particles, no dark energy fluid. The entire cosmic history—from the Big Bang singularity to the present accelerated expansion—is governed by a single geometric degree of freedom (the R^2 scalaron) whose behavior transitions from UV renormalization group dynamics to IR saturation dynamics.

B. Implications for Paper I: The Nash equilibrium in the ultraviolet

The UV completion has direct consequences for the game-theoretic framework of Paper I [1], which modeled cosmic evolution as a Nash equilibrium between a metastable quantum vacuum ("null space") and a spacetime bubble. We can now assign physical identities to both players.

a. Player identification. The null space of Paper I corresponds to the pure R^2 regime of QQG: a scale-invariant theory without classical spacetime, where the universe exists as a quantum state near the Gaussian UV fixed point. The spacetime bubble corresponds to the emergent GR regime after the strong-coupling transition at $\mu \sim \mu_0$. The Nash equilibrium—the endpoint of the game—maps to the tanh saturation attractor, which the Saturation Theorem guarantees as the unique stable outcome.

b. The complete five-phase game. Paper I described only the late-time phases (geometric dark matter → saturation). QG-CRM extends the game to all five phases:

1. *Inflation*: the null-space player dominates (pure R^2, no GR).

2. *Kination/reheating*: the "move" passes from player 1 to player 2 (strong-coupling transition, GR emerges).

3. *Radiation/matter*: the spacetime-bubble player dominates (standard cosmology).

4. *Geometric dark matter*: cooperative phase (both players contribute).

5. *Saturation*: Nash equilibrium is reached (de Sitter end state).

c. Parameter derivation: what the UV fixes and what it does not. In Paper I, the saturation amplitude Φ_0 and relaxation rate k were fitted to Pantheon+ data. The UV completion clarifies the *logical status* of these parameters: QQG determines (i) the *existence* of a finite saturation value (via $\xi_{\max}$), (ii) the *functional form* of relaxation (tanh, guaranteed by the Saturation Theorem), and (iii) the *initial conditions* for the saturation ODE at the reheating surface. However, the *numerical values* of Φ_0 and k are IR quantities, determined jointly by the UV initial conditions and the cosmological boundary conditions (H_0, Ω_m). The strong-coupling transition is the correct UV-IR matching surface, but it transfers the dynamics to the IR regime rather than fixing Φ_0 directly. A fully quantitative derivation would require tracking the RG flow through the nonperturbative strong-coupling window—an open problem analogous to computing hadron masses from the QCD Lagrangian.

d. Potential identification: projection, not identity. The game-theoretic potential of Paper I and the QQG Einstein-frame potential (8) are not the same object, but represent *different levels of description of the same system.* The QQG potential is a microscopic RG-improved potential, valid in the UV/inflationary regime. The Paper I potential is an effective IR free-energy potential, defined through relaxation dynamics and thermodynamic selection. The connection between them is mediated by the Saturation Theorem: the QQG potential provides the UV initial conditions, and the tanh normal form projects these onto the unique IR attractor. In Lagrangian language, the Paper I potential corresponds to the effective potential of the scalaron in the Einstein frame, evaluated at cosmological densities and integrated over the full RG flow including the strong-coupling transition.

e. Thermodynamic strengthening. Paper I established a thermodynamic equivalence connecting the game-theoretic axioms to the Jacobson tradition [28]. The UV completion strengthens this connection through three independent results: (i) the quantum-thermodynamic de Sitter decay mechanism [26, 27], (ii) Mottola's thermodynamic instability of de Sitter space [19, 20], and (iii) the quantum break-time of de Sitter [18]. All three provide concrete physical mechanisms for the "game dynamics" that Paper I described axiomatically.

C. Comparison with alternative frameworks

TABLE II. Comparison of cosmological frameworks.

	ΛCDM	Starobinsky	QG-CRM
Inflaton field	Yes	Yes (scalaron)	No
Cosmological constant	Yes	No	No
Dark matter particles	Yes	Yes	No
UV complete	No	No	Yes†
n_s prediction	—	$1-2/N$	$1-4/(3N)$
r prediction	—	$12/N^2$	$\gtrsim 0.01$
MOND emergence	No	No	Yes

† Conditional on ghost confinement in the strong-coupling regime (see Sec. VIII E).

D. Robustness of the large-$\mathcal{N}$ requirement

The viable parameter space requires $\mathcal{N} \sim 10^5$–10^6 matter fields contributing to the QQG beta functions. We identify three classes of beyond-standard-model (BSM) scenarios that naturally provide this:

1. **Large gauge groups.** An SU(N_c) theory with $N_c \sim 300$–1000 and a small number of fundamental fermions yields $\mathcal{N} \sim N_c^2/5 \sim 10^4$–$2\times10^5$ from the adjoint gauge bosons alone. This is the scenario most closely related to holographic cosmology [33], where N_c^2 plays the role of the central charge.

2. **Large hidden sectors.** String landscape constructions with $\sim 10^5$ SM-like gauge sectors yield $\mathcal{N} \sim 5\times10^5$ while preserving individual sector simplicity.

3. **Scalar moduli.** Type IIB flux compactifications with $\sim 6\times10^6$ scalar moduli give $\mathcal{N} \sim 10^5$ from the scalar contribution alone ($\mathcal{N}_{\rm scalar} = N_{\rm scalar}/60$).

We stress the distinction between *existential* and *realistic* large $\mathcal{N}$: the QG-CRM framework *requires* $\mathcal{N} \sim 10^5$–10^6 for perturbative control, but it does not specify which fields provide this. The scenarios above demonstrate that such field content is *available* in well-motivated BSM settings, not that it is uniquely determined.

Crucially, these fields do *not* contribute to the effective number of relativistic species $N_{\rm eff}$ at BBN or CMB decoupling. This is a structural feature, not a fine-tuning: in QG-CRM, the large-$\mathcal{N}$ fields are active *exclusively* above the strong-coupling scale ($\mu > \mu_0$). At the confinement transition, they are integrated out together with the QQG degrees of freedom—precisely as heavy quarks are integrated out above their mass threshold in QCD. Below the confinement scale, only standard model fields ($\mathcal{N}_{\rm SM} \approx 4.7$) propagate as asymptotic states, yielding $\Delta N_{\rm eff} = 0$ from the hidden sector. Entropy production at reheating is $S_f/S_i \sim 1$–3, consistent with standard baryogenesis.

E. Ghost confinement in the Weyl sector

The C^2 term in quadratic gravity introduces a massive spin-2 field with negative kinetic energy—a "ghost" in the perturbative spectrum. However, this does not imply the existence of propagating ghost *asymptotic states.*

In QG-CRM, the ghost is a *confined resonance*, not an on-shell particle: as λ grows toward the strong-coupling regime ($\lambda_{\rm tH} \gtrsim 1$), the ghost becomes confined, in direct analogy to how gluons in QCD appear in the perturbative spectrum but never as free asymptotic states [14]. Three aspects support this:

1. **Asymptotic freedom.** Both QQG and QCD are asymptotically free, with perturbative UV behavior and nonperturbative IR dynamics. In QCD, confinement prevents free quarks from appearing as asymptotic states; in QQG, confinement is expected to prevent the ghost from appearing as a physical excitation.

2. **Dynamical mass generation.** In the strong-coupling regime, the ghost may acquire a dynamical mass $m_{\rm ghost} \sim \mu_0$, pushing its effects above the energy scale of cosmological observations. This is analogous to constituent quark masses in QCD.

3. **Observable signatures.** If ghost confinement is incomplete, the most likely signature would be a modification of the tensor power spectrum at scales $k \sim m_{\rm ghost}$, producing a feature in the B-mode polarization at high ℓ. The absence of such features in current CMB data provides an indirect consistency check, though dedicated searches with Stage IV experiments would be valuable.

We acknowledge that a rigorous proof of ghost confinement in QQG remains outstanding. Lattice formulations of quadratic gravity [14] offer the most promising path toward a nonperturbative verification.

F. Nonperturbative scenarios for Φ_0 and k

While the precise numerical values of the CRM parameters Φ_0 and k require nonperturbative UV-IR matching, we can sketch two plausible scenarios:

a. Scenario A: Dimensional transmutation. By analogy with $\Lambda_{\rm QCD}$, the strong-coupling transition in QQG generates a dynamical scale $\mu_{\rm conf} \sim \mu_0\, e^{-1/\lambda_{\rm tH}}$. The ratio $\mu_{\rm conf}/M_{\rm Pl}$ then sets the vacuum energy scale, and the CRM parameters inherit their values from this dynamically generated hierarchy. In this scenario, $\Phi_0 \propto (\mu_{\rm conf}/M_{\rm Pl})^4$ and k is set by the logarithmic derivative of ξ at the confinement scale.

b. Scenario B: Thermodynamic selection. The FST dark energy paper [15] derives $\Lambda_{\rm eff} \sim H_0^2/\ln(M_{\rm Pl}/H_0)$ from a thermodynamic variational principle. If the QQG RG flow provides the microscopic underpinning for this variational principle—with the free-energy functional $\Phi(\rho_{\rm vac})$ emerging from the path integral of QQG in the strong-coupling regime—then Φ_0 is selected by thermodynamic equilibrium rather than by UV dynamics alone.

Both scenarios predict $\Phi_0 \sim \mathcal{O}(1)$ (in units of the critical density), consistent with the observed $\Omega_\Lambda \approx 0.685$, but a derivation from first principles remains a major open challenge.

G. Perturbative stability: two-loop estimate

We estimate the impact of 2-loop corrections on the inflationary predictions using the parametric scaling $\delta(\text{obs})/\text{obs} \sim \lambda_{\rm tH}/(4\pi)^2$. For $\lambda_{\rm tH} = 1$ (the strong-coupling boundary), this gives $\delta \sim 0.6\%$. Concretely:

- The spectral index shift is $|\delta n_s| < 1.5\times10^{-4}$, which is $20\times$ smaller than the current 1σ observational error ($\sigma_{n_s} \approx 0.003$). The n_s prediction is *robust*.

- The tensor-to-scalar ratio shift is $|\delta r/r| < 0.6\%$, corresponding to $|\delta r| < 10^{-4}$. This is below LiteBIRD's forecast sensitivity ($\sigma_r \sim 10^{-3}$), though marginally detectable at the $\sim 10\%$-of-σ level.

- Perturbative control breaks down only at $\lambda_{\rm tH} \gtrsim 2$, well outside the viable parameter space ($0.1 \lesssim \lambda_{\rm tH} \lesssim 1$).

We conclude that the 1-loop predictions of Table I are stable against higher-order corrections throughout the viable parameter space.

H. Remaining open questions

1. **$\mathbf{R}^2 \to$ nonlinear $f(R)$ transition** [*requires nonperturbative methods*]. The Convexity-Screening Theorem mandates nonlinear $f(R)$ in the IR, but the confinement mechanism that generates Hu-Sawicki structure from the UV R^2 action is inherently nonperturbative. Lattice methods for quadratic gravity may provide a path forward.

2. **RG matching for $\Lambda_{\rm eff}$** [*plausible, open*]. The missing factor $\sim 290 \approx 2\ln(M_{\rm Pl}/H_0)$ in the thermodynamic bound is suggestive of a 2-loop $\ln^2$ contribution or large-$\mathcal{N}$ enhancement, but the calculation remains open.

3. **Perturbations from the Weyl sector.** Running of the C^2 coupling may modify the tensor power spectrum and hence r; a quantitative assessment is needed.

These are genuine *research questions*, not structural deficiencies. The QG-CRM framework is internally consistent at the level presented; the open questions concern quantitative refinements within a fixed architecture.

IX. CONCLUSION

We have presented QG-CRM, the ultraviolet-completed Curvature Relaxation Model. By identifying the R^2 sector of the CRM Lagrangian with asymptotically free quantum quadratic gravity, and by verifying that the Saturation Theorem provides the unique UV-IR interface, we obtain a complete geometric cosmology spanning from the Big Bang to the present epoch.

The key results are:

1. The UV completion is provided by quantum quadratic gravity: renormalization group running of the R^2 coupling dynamically generates inflation without an inflaton field.

2. The Saturation Theorem (Paper V) applies to the QQG RG flow, guaranteeing tanh as the unique IR normal form and establishing the CRM saturation profile as structurally necessary.

3. The UV completion answers the open question of Paper V by identifying QQG as the theory that selects the CRM parameters: it determines the existence and functional form of saturation, fixes the inflationary observables, and provides scaling relations for the IR parameters Φ_0 and k—though their precise numerical values require nonperturbative UV-IR matching.

4. QG-CRM makes falsifiable predictions—$n_s \sim 1 - 4/(3N)$, $r \gtrsim 0.01$, phantom-like $w(z)$, and $a_0 = cH_0/(2\pi)$—that are testable with Stage IV CMB experiments, gravitational wave detectors, and galaxy surveys.

QG-CRM is, to our knowledge, the first framework that unifies ultraviolet-complete inflation, geometric dark matter, MOND dynamics, and late-time cosmic acceleration within a single action principle—using only spacetime curvature and baryonic matter.

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Appendix A: Explicit Parameter Mapping: QQG → CRM

The parameter mapping must be understood at two levels: what the UV completion *determines structurally* and what it *leaves to IR boundary conditions*.

a. What QQG determines.

1. **Existence of saturation.** The RG trajectory has a finite $\xi_{\max}$, guaranteeing that the relaxation variable is bounded. This is Axiom A of the Saturation Theorem.

2. **Functional form.** The Saturation Theorem forces tanh as the unique normal form. No alternative profile is permitted.

3. **Initial conditions.** At the reheating surface ($\mu \sim \mu_0$, where strong coupling sets in and GR emerges), the QQG dynamics sets the initial value and velocity of the relaxation variable.

4. **Inflationary observables.** n_s, r, and A_s are fully determined by $\lambda_{\rm tH}$ and N (Table I).

b. What requires IR boundary conditions. The saturation amplitude Φ_0 and relaxation rate k are IR quantities. Their numerical values are determined jointly by:

- the UV initial conditions at the reheating surface,
- the cosmological expansion history through radiation and matter domination,
- the present-day boundary conditions (H_0, Ω_m).

c. UV-derived scaling relations. While Φ_0 and k cannot be computed from UV parameters alone, the UV completion provides *scaling relations*:

- The inflation energy scale sets an upper bound:

$$\frac{V_{\rm inf}}{M_{\rm Pl}^4} = \frac{35\lambda_0^2\mu_0^4}{128\pi^2\lambda_{\rm tH}\, M_{\rm Pl}^4} \sim 10^{-9}, \tag{A1}$$

  confirming that the inflationary vacuum energy is far above the present dark energy scale.

- The RG slope at the tachyon divide sets the *rate* at which the scalaron coupling evolves:

$$\left.\frac{d\xi}{d\ln\mu}\right|_{\mu=\mu_0} = \frac{2520\lambda_0^2}{36(4\pi)^2} \sim 4.4\times 10^{-5}, \tag{A2}$$

  providing a characteristic timescale for the UV→IR transition.

- **The relaxation rate k as a critical exponent.** At the tachyon divide ($\xi = 0$), the linearized RG flow of ξ has the Jacobian

$$\left.\frac{\partial\beta_\xi}{\partial\xi}\right|_{\xi=0} = \frac{\lambda_0}{(4\pi)^2}. \tag{A3}$$

  This quantity plays the role of a *critical exponent*: it characterizes the rate at which the system departs from the critical surface $\xi = 0$, independently

of the nonperturbative dynamics beyond this surface. The CRM relaxation rate k inherits this universal scaling:

$$k \propto \frac{\lambda_0}{(4\pi)^2} \sim 7 \times 10^{-7}, \tag{A4}$$

where the numerical value uses $\lambda_0 \sim 10^{-4}$ from the A_s constraint. This is a structural result: k is determined by the *local* properties of the RG flow at the critical surface, not by the global nonperturbative dynamics.

Important clarification: The RG exponent $k_{\rm RG} \sim 7 \times 10^{-7}$ and the observed CRM relaxation rate $\kappa = 2.63$ are *not the same quantity.* They differ by the Jacobian of the coordinate transformation between RG time ($\ln \mu$) and cosmic time ($\ln a$):

$$\kappa = k_{\rm RG} \cdot \left| \frac{d \ln \mu}{d \ln a} \right| \cdot \mathcal{J}_{\rm repar}, \tag{A5}$$

where $\mathcal{J}_{\rm repar}$ is the reparametrization Jacobian from the Saturation Theorem (which guarantees tanh "up to reparametrization"). Since $\mu \sim |R|^{1/2}$ and $R \propto H^2 \propto a^{-3}$ during matter domination, $|d \ln \mu / d \ln a| = 3/2$ per e-fold of expansion. Over the full cosmic evolution from reheating ($a \sim 10^{-28}$) to present ($a = 1$), the integrated Jacobian spans ~ 64 e-folds of expansion, each contributing $\mathcal{O}(1)$ to the mapping. The remaining factor $\sim 10^{4-5}$ arises from the reparametrization between the RG control variable and the CRM relaxation variable—a nontrivial but structurally expected consequence of projecting a UV flow onto an IR normal form.

- The reheating temperature from out-of-equilibrium dS decay [6]:

$$T_{\rm RH} = 10^{-10} \left(\frac{E_{\rm inf}}{\rm GeV} \right)^{3/2} \ {\rm GeV} \sim 10^{11}\text{–}10^{12}\,{\rm GeV}, \tag{A6}$$

consistent with baryogenesis.

d. The nonperturbative gap. A fully quantitative derivation of Φ_0 and k from UV parameters would require solving the RG flow through the strong-coupling regime at $\lambda_{\rm tH} \sim 1$. This is analogous to computing hadron masses from the QCD Lagrangian: the theoretical framework is established, but the nonperturbative computation remains an open challenge. Lattice methods for quadratic gravity [14] may ultimately provide the necessary bridge.

Appendix B: Scalar Perturbation Power Spectrum

The primordial scalar power spectrum in QG-CRM has two contributions:

1. **Primary (scalaron):** The standard $f(R)$-inflation result from the Einstein-frame potential (8), yielding [6]

$$\Delta^2_\phi(k) \propto \frac{\lambda_0^2 N^{4/3}}{(2\lambda_{\rm tH})^{1/3}}. \tag{B1}$$

2. **Secondary (tensor-sourced):** The second-order contribution from tensor perturbations via the kernel computed in [7]. This is suppressed relative to the scalaron channel by a factor of order $(H_{\rm inf}/m_{\rm pl})^2 \ll 1$ and is therefore subleading.

The detailed computation of the combined spectrum, including interference effects and non-Gaussian corrections, is deferred to future work.

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